

Diabetic Data Predictive Analysis

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Contents

Variable Exploration	2
Number of Encounters	2
Patient Age	3
Discharge Disposition ID	4
Diabetes Medication	5
Regressions	6
Age and Medication	6
Discharge Description and Number of Encounters	6
Assessment and Conclusions	7

Variable Exploration

Number of Encounters

nbr_encounters – The number of encounter IDs associated with each patient number.
 Calculated using a countif() statement to count the number of times each patient number appears.

nbr_encounters	
Max	40
Median	1
Min	1
Average	2.259025608

	nbr_encounters	30readm
nbr_encounters	1	
30readm	0.236953388	1

Row Labels	Average of 30readm	Count of Partition
1	0.040260689	32836
2	0.145306643	12539
3	0.191553043	6014
4	0.210758377	3402
5	0.228718428	2138
6	0.24453125	1280
7	0.266119578	853
8	0.270058708	511
9	0.29280397	403
10	0.373015873	252
11	0.417322835	127
12	0.318181818	132
13	0.435643564	101
14	0.355555556	45
15	0.36	75
16	0.4375	32
17	0.5	32
18	0.459016393	61
19	0.441176471	34
20	0.473684211	76
21	0.461538462	13
22	0.454545455	22
23	0.461538462	39
28	0.6875	16
40	0.538461538	26
Grand Total	0.110417793	61059

Patient Age

age – age of each patient grouped in 10 year intervals from 0 to 100

age	count
[0-10)	161
[10-20)	691
[20-30)	1657
[30-40)	3775
[40-50)	9685
[50-60)	17256
[60-70)	22483
[70-80)	26068
[80-90)	17197
[90-100)	2793

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
[0-10)	97	2	0.020618557	0.02040378		
[10-20)	426	25	0.058685446	0.055371444		
[20-30)	1005	140	0.139303483	0.120017443		
[30-40)	2287	272	0.1189331	0.104833857		
[40-50)	5833	617	0.105777473	0.094604818		
[50-60)	10363	1000	0.096497153	0.087193867		
[60-70)	13377	1473	0.110114375	0.097996525		
[70-80)	15596	1802	0.115542447	0.102198943		
[80-90)	10395	1219	0.117267917	0.103526112		
[90-100)	1680	192	0.114285714	0.101284778		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	5.984172737	9	0.664908082	6.774837321	8.80528E-10	1.880039123
Within Groups	5991.579069	61049	0.098143771			
Total	5997.563242	61058				

Row Labels	Average of 30readm	Count of Partition
[0-10)	0.020618557	97
[10-20)	0.058685446	426
[20-30)	0.139303483	1005
[30-40)	0.1189331	2287
[40-50)	0.105777473	5833
[50-60)	0.096497153	10363
[60-70)	0.110114375	13377
[70-80)	0.115542447	15596
[80-90)	0.117267917	10395
[90-100)	0.114285714	1680
Grand Total	0.110417793	61059

Discharge Disposition ID

discharge_disposition_id – Integer identifier corresponding to unique discharge descriptions. The ten most common discharges were used for this analysis.

discharge_disposition_id	count
1	60234
2	2128
3	13954
4	815
5	1184
6	12902
11	1642
18	3691
22	1993
25	989

discharge_disposition_id	description
1	Discharged to home
2	Discharged/transferred to another short term hospital
3	Discharged/transferred to SNF
4	Discharged/transferred to ICF
5	Discharged/transferred to another type of inpatient care institution
6	Discharged/transferred to home with home health service
11	Expired
18	NULL
22	Discharged/transferred to another rehab fac including rehab units of a hospital .
25	Not Mapped

Row Labels	Average of 30readm	Count of Partition
1	0.090914117	36177
2	0.1616	1250
3	0.150202816	8382
4	0.115768463	501
5	0.202216066	722
6	0.123427979	7713
11	0	991
18	0.125338142	2218
22	0.278290025	1193
25	0.098305085	590
Grand Total	0.110049048	59737

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
1	36177	3289	0.090914117	0.082651025		
2	1250	202	0.1616	0.135593915		
3	8382	1259	0.150202816	0.12765716		
4	501	58	0.115768463	0.102570858		
5	722	146	0.202216066	0.16154848		
6	7713	952	0.123427979	0.108207542		
11	991	0	0	0		
18	2218	278	0.125338142	0.109677942		
22	1193	332	0.278290025	0.201013181		
25	590	58	0.098305085	0.088791689		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	83.98217011	9	9.331352235	96.64932314	4.2505E-180	1.880042503
Within Groups	5766.555386	59727	0.096548552			
Total	5850.537556	59736				

Diabetes Medication

diabetesMed – Indicates whether diabetes medication was prescribed with values of “Yes” or “No”

diabetesMed	count
Yes	78363
No	23403

Row Labels	Average of 30readm	Count of Partition
Yes	0.115471377	46990
No	0.093538986	14069
Grand Total	0.110417793	61059

Anova: Single Factor						
SUMMARY						
<i>Groups</i>	<i>Count</i>	<i>Sum</i>	<i>Average</i>	<i>Variance</i>		
0	14069	1316	0.093538986	0.084795472		
1	46990	5426	0.115471377	0.102139912		
ANOVA						
<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
Between Groups	5.208239242	1	5.208239242	53.06752741	3.26152E-13	3.841611128
Within Groups	5992.355003	61057	0.09814362			
Total	5997.563242	61058				

Regressions

Age and Medication

SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.043584063								
R Square	0.001899571								
Adjusted R Square	0.001736076								
Standard Error	0.313140196								
Observations	61059								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	10	11.39279473	1.139279473	11.61856881	3.14851E-20				
Residual	61048	5986.170447	0.098056782						
Total	61058	5997.563242							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	0.120517722	0.002595395	46.43521554	0	0.11543074	0.125604703	0.11543074	0.125604703	
[0-10)	-0.095516552	0.03189339	-2.994869844	0.00274672	-0.158027688	-0.033005417	-0.158027688	-0.033005417	
[10-20)	-0.058575909	0.015379243	-3.808764157	0.0001398	-0.088719269	-0.028432549	-0.088719269	-0.028432549	
[20-30)	0.023438744	0.010191081	2.299927217	0.021455701	0.003464197	0.043413291	0.003464197	0.043413291	
[30-40)	0.003854889	0.007011914	0.549762788	0.582484096	-0.009888482	0.01759826	-0.009888482	0.01759826	
[40-50)	-0.009573387	0.004806107	-1.991921356	0.046384134	-0.01899337	-0.000153404	-0.01899337	-0.000153404	
[50-60)	-0.01902881	0.003968565	-4.794884236	1.63152E-06	-0.026807209	-0.011250411	-0.026807209	-0.011250411	
[60-70)	-0.005465827	0.003690201	-1.481173234	0.138565578	-0.01269863	0.001766977	-0.01269863	0.001766977	
[80-90)	0.002383075	0.003965881	0.600894146	0.547912724	-0.005390064	0.010156214	-0.005390064	0.010156214	
[90-100)	0.000786626	0.008045504	0.097772108	0.922113577	-0.014982585	0.016555836	-0.014982585	0.016555836	
No	-0.02237439	0.003012636	-7.426847321	1.12661E-13	-0.028279166	-0.016469615	-0.028279166	-0.016469615	

Discharge Description and Number of Encounters

SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.262408435								
R Square	0.068858187								
Adjusted R Square	0.06870566								
Standard Error	0.302454195								
Observations	61059								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	10	412.9813287	41.29813287	451.4515956	0				
Residual	61048	5584.581913	0.09147854						
Total	61058	5997.563242							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	0.025893713	0.001912267	13.5408451	1.03123E-41	0.022145664	0.029641761	0.022145664	0.029641761	
2	0.067587084	0.008696163	7.772058307	7.84211E-15	0.05054258	0.084631587	0.05054258	0.084631587	
3	0.052787241	0.003655236	14.4415428	3.39126E-47	0.045622969	0.059951513	0.045622969	0.059951513	
4	0.017622911	0.013602986	1.29551787	0.195146539	-0.009038979	0.044284801	-0.009038979	0.044284801	
5	0.116956329	0.01136461	10.29127477	8.08401E-25	0.09468166	0.139230997	0.09468166	0.139230997	
6	0.020339023	0.003785856	5.372371256	7.79916E-08	0.012918735	0.027759311	0.012918735	0.027759311	
11	-0.078101218	0.009736724	-8.021303843	1.06472E-15	-0.097185224	-0.059017212	-0.097185224	-0.059017212	
18	0.031254734	0.006609402	4.728829426	2.26322E-06	0.018300288	0.04420918	0.018300288	0.04420918	
22	0.187294016	0.008894903	21.05633026	4.47419E-98	0.169859981	0.204728052	0.169859981	0.204728052	
25	-0.001930806	0.012550131	-0.153847456	0.877730524	-0.026529099	0.022667488	-0.026529099	0.022667488	
nbr_encounters	0.030291357	0.000504114	60.08824845	0	0.029303291	0.031279423	0.029303291	0.031279423	

Assessment and Conclusions

The regression with discharge description and the number of encounters was the better predictor of 30 day readmissions. The regression ran on discharge description and the number of encounters had a higher R Square value than the regression ran on age and medication. The regression ran on discharge description and the number of encounters also had a slightly lower average squared error when predicting 30 day readmissions in both the training and testing data sets.

Patients discharged to another short term hospital have a 6.76% greater probability of being readmitted within thirty days than patients discharged to home, and the difference is statistically significant. Patients discharged to skilled nursing facility (SNF) have a 5.28% greater probability of being readmitted within thirty days than patients discharged to home, and the difference is statistically significant. Patients discharged to intermediate care facility (ICF) have a 1.76% greater probability of being readmitted within thirty days than patients discharged to home, and the difference is not statistically significant. Patients discharged to another type of inpatient care institution have an 11.7% greater probability of being readmitted within thirty days than patients discharged to home, and the difference is statistically significant. Patients discharged to home with home health service have a 2.03% greater probability of being readmitted within thirty days than patients discharged to home, and the difference is statistically significant. Number 11 in the discharge codes means “expired” or that the patient has died. This of course means that the patient will not be readmitted. The interpretation is not likely to be useful, but patients that expired have a 7.81% lower probability of being readmitted than patients that were discharged to home in the model. The difference is statistically significant. Patients who have a null value for discharge disposition have a 3.13 % greater probability of being readmitted within thirty days than patients discharged to home, and the difference is statistically significant. Patients discharged to another rehab facility including rehab units of a hospital have an 18.73% greater probability of being readmitted within thirty days than patients discharged to home, and the difference is statistically significant. Patients whose discharge disposition was not mapped have a 0.19% lower probability of being readmitted within thirty days than patients discharged to home, and the difference is not statistically significant. For each additional encounter a patient has, the probability of being readmitted within 30 days increases by 3.03%. This figure is statistically significant. The whole model explains approximately 6.89% of the variation in the 30 day readmission rate. Overall, the model is statistically significant with a p-value that rounds to zero.

According to the model, patients who have already made multiple visits are more likely to be readmitted within 30 days. It may be a good idea to place an emphasis on the

care of these patients. The differences in discharge description may not be as relevant for decision making. For example, patients who were discharged to a rehab facility are the most likely group in the model to be readmitted, but the patients who were discharged to a rehab facility may not have been in good enough condition to be discharged home. In this case, it would not make sense to send a patient home due to the lower probability of being readmitted within 30 days. However, it may be useful in predicting the volume of 30 day readmissions within a certain time frame. For example, if several patients are discharged to a rehab facility, we may predict that there will be more readmissions than there would have been otherwise.